Revised Full Mathematical Peer Review of the 108-Page S6 Manuscript

Corrective addendum, proof-status reconstruction, and Codex-finding integration

Independent technical review

25 August 2026

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1 Corrective addendum dated 25 August 2026

This edition integrates the mathematical claims contained in the user-supplied Codex progress log, but it does not treat that log as evidence by itself. The log reports two strengthened proof lines: a direct period-theoretic proof that

$$a(X) = 1, \qquad \mathcal{M}(X) = f^*\mathbb{C}(t),$$

and an integral nearby-cycle/Leray computation that proves the homology-sphere conclusion without the global cusp-collapse theorem. It also reports corrections to a modular-frame factor, a Neron–Severi simplification, Lemma A.3's variance, and the distinction between Λ and $\bar{\Lambda}$. Only the progress log, not the separate Codex workbench files or its claimed 38-check harness, was supplied. Every point promoted in this report was therefore re-located and re-checked in the audited 108-page manuscript itself.

The result is a genuine but limited correction to the review of 24 August 2026.

1. The direct algebraic-dimension argument in Section 9.1 is stronger than the earlier review stated. Its period calculation, very-general Neron–Severi computation, exclusion of algebraic dimensions one and two for the fibres, and meromorphic descent form a complete argument conditional only on the construction of the period family and standard structure theory for complex tori. It is independent of Section 7 and of Campana–Demailly–Peternell.
2. The second topology route in Section 7.7 and Appendix B is not merely corroborative. It is a logically sufficient topology engine for the recognition of X as S^6, provided the based fundamental-group relations, the finite-fibre specialization lattices, the integral cusp specialization theorem, and the geometric identification of the Leray transgressions are correct. Proposition 7.2's global collapse is therefore not theorem-critical for the sphere-recognition chain, although it remains necessary for the manuscript's first topology route and for its explicit collapse-map and generator claims.

3. Lemma A.3 contains a confirmed variance error. The displayed Mayer–Vietoris sequence and every map subsequently computed are cohomological, although the proof writes H_k. Replacing H_k by H^k throughout that calculation and applying universal coefficients at the end repairs the lemma without changing its stated groups.
4. The corrected source already contains the modular factor $\mathsf{e} = F^{-1}e$ needed on source page 71, and the current algebra in Lemma 9.3 is correct. Those are repairs present in the audited text, not defects of the current version.
5. The current cusp notation correctly uses $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$ as the deck group. The Codex log does not identify the earlier erroneous occurrence, so no additional current-source error is asserted on that point.

The recommendation remains **major revision and independent specialist replication**. The reason is no longer a suspicion that the global collapse is the sole route to the homology calculation, nor an unresolved algebraic-dimension argument. The remaining burden is concentrated in the actual geometric realization of the displayed local and global maps, integral nearby-cycle functoriality and orientation data, the based van Kampen and transgression comparisons, and the analytic duality/base-change/literature-comparison chain in Section 10.

1.1 Mathematical results so far: a cold-entry guide

1.1.1 The construction being claimed

The manuscript starts with a rank-four integral lattice Λ and monodromies of orbifold orders 3 and 4, together with a unipotent cusp monodromy. Period functions τ, μ, β define a holomorphic family of two-dimensional complex tori over the thrice-punctured projective line. Three local fillings are then attached: a toric, reduced, irreducible, nonnormal normal-crossing fibre W at the cusp, and multiple fibres $3S_1$ and $4S_2$ whose reductions are bielliptic surfaces. The gluing parameters enter the topology only through

$$p = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

For the actual data $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, one has $p = -1$.

1.1.2 The topology chain now isolated

The based van Kampen calculation produces

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \rangle \cong \mathbb{Z}/|p|,$$

conditional on the geometric identification of the three meridians and their drift terms. Thus the actual value $p = -1$ gives $\pi_1(X) = 1$.

The independent Leray route uses the exact cusp-specialization statement

$$H^q(W; \mathbb{Z}) \xrightarrow{\mathrm{sp}^q} H^q(F; \mathbb{Z})^{T_0}$$

as an isomorphism for $0 \le q \le 4$, together with the finite-fibre specialization indices. It obtains

$$H^0(B, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\gamma, \quad H^0(B, R^2 f_* \mathbb{Z}) = \mathbb{Z} \cdot 2q, \quad H^0(B, R^3 f_* \mathbb{Z}) = \mathbb{Z} \cdot 2\gamma uw.$$

The three relevant transgressions are

$$\begin{aligned} d_2^{0,1}(12\gamma) &= \pm p\, \omega, \\ d_2^{0,2}(2q) &= \pm p\, \omega \bar{\delta}, \\ d_2^{0,3}(2\gamma uw) &= \pm p\, \omega [\gamma \delta], \end{aligned}$$

with one common sign. Multiplicativity gives the same integer in all three formulas. When $p = -1$, these maps are units. The spectral sequence then yields $H^1(X) = H^2(X) = H^3(X) = 0$; universal coefficients and Poincare duality give the integral homology of S^6. Together with $\pi_1(X) = 1$ and $\Theta_6 = 0$, this implies $X \cong_{\mathrm{diff}} S^6$.

This route does not use the global collapse theorem. Its theorem-critical unresolved inputs are instead: the based meridian relations, the integral specialization theorem at W, the finite-fibre lattices, and the geometric realization of the first transgression and its multiplicative comparison.

1.1.3 The direct algebraic-dimension chain

For an integral class

$$E = E_{\gamma u}\gamma u + E_{\gamma w}\gamma w + E_{\gamma\delta}\gamma\delta + E_{uw}uw + E_{u\delta}u\delta + E_{w\delta}w\delta,$$

Section 9.1 proves

$$E \in \mathrm{NS}(F_b) \quad\Longleftrightarrow\quad P_E(z) = 0,$$

where

$$P_E = E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta + E_{w\delta}(6\mu^2 - \tau\beta).$$

The transformation laws under the two orbifold generators prove that

$$1,\ \tau,\ \mu,\ \beta,\ 6\mu^2 - \tau\beta$$

are linearly independent. Hence the very general fibre has

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad \eta = uw + 6\gamma\delta.$$

Moreover

$$\eta^2 = 12, \qquad G_\eta = \begin{pmatrix} 2\,\mathrm{Im}\,\tau & 2\,\mathrm{Im}\,\mu \\ 2\,\mathrm{Im}\,\mu & \frac{1}{3}\,\mathrm{Im}\,\beta \end{pmatrix},$$

and

$$\det G_\eta = \frac{2}{3}\,\mathrm{Im}\,\tau\left(\mathrm{Im}\,\beta - \frac{6(\mathrm{Im}\,\mu)^2}{\mathrm{Im}\,\tau}\right) < 0.$$

Thus every nonzero multiple of η is nondegenerate and indefinite, while its square is nonzero. A two-torus of algebraic dimension two would have an ample positive class; one of algebraic dimension one would have a nonzero fibre class of square zero. Neither can occur when $\mathrm{NS} = \mathbb{Z}\eta$. Therefore the very general fibre has algebraic dimension zero.

A separate descent lemma shows that a meromorphic function on X which is constant on uncountably many smooth fibres comes from the base. Since very general fibres have algebraic dimension zero,

$$\mathcal{M}(X) = f^*\mathcal{M}(\mathbb{P}^1) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction. This proof is independent of the topology and of the CDP dispute.

1.1.4 What Section 10 has and has not established

The local conductor calculation produces a nonzero ambient cotangent section on a reduced nonnormal normal-crossing fibre even when the torsion-free intrinsic cotangent sheaf has no section. For a line bundle A trivial after normalization, the pair

$$\big(\eta^*(df|_W)\otimes e, 0\big)$$

descends through the normalization/conductor sequence to

$$s \in H^0(W, \Omega^1_X|_W \otimes A),$$

whose intrinsic image is conductor-supported torsion. This local mechanism is mathematically real and was independently reconstructed.

Its application to the global contradiction requires the exact analytic chain

$$H^2(W,E)^\vee \simeq \mathrm{Hom}_W(E,\omega_W) \simeq H^0(W, E^\vee\otimes\omega_W), \qquad E=(T_X\otimes L)|_W,$$

followed by the top-degree base-change map

$$(R^2 f_*(T_X\otimes L))_0 \otimes k(0) \longrightarrow H^2(W,(T_X\otimes L)|_W).$$

The manuscript states the relevant Gorenstein duality and top-degree base-change hypotheses, and no internal contradiction was found. Nevertheless, three assertions must remain distinct:

1. the local computation may expose a gap in a printed proof at a nonnormal fibre;
2. that does not by itself prove the published theorem false in all stated generality;
3. the theorem is contradicted by this example only after the construction and every hypothesis on X have independently been certified.

1.2 Revised status ledger

Status	Claim	Present assessment
VERIFIED exact algebra	monodromy, Smith forms, invariant/coinvariant lattices, period transformations, P_E, η^2, Gram determinant, Appendix B rank matrices	Reproduced in two local harnesses: 28 legacy checks and 14 supplemental checks, all passing.
CONDITIONALLY VALIDATED	direct proof of $a(X)=1$ and $\mathcal{M}(X)=f^*\mathbb{C}(t)$	The complete argument is present and survives line-by-line review, conditional on the period-family construction and standard complex-torus structure theory.
CONDITIONALLY VALIDATED	Leray route from specialization and transgression data to $H_*(X)=H_*(S^6)$	Formal sheaf, spectral-sequence, rank, saturation, and multiplicativity logic survives review. Geometric bridge inputs remain open to specialist replication.
OPEN	existence and exact gluing of the global compact complex threefold	No fatal defect found, but the infinite toric quotient, finite logarithmic fillings, and Hausdorff/proper gluing remain construction-level obligations.

Status	Claim	Present assessment
OPEN	Section 10 as a counterexample to the printed CDP conclusions	Local conductor mechanism is strong; global duality/base change and exact comparison with the published proof require specialist adjudication.
REFUTED local wording	Remark 7.28 as a fixed-triple biconditional	False; must be existential in ℓ_0.
REFUTED proof sentence	Proposition 4.7(iv)'s properness justification	Closedness needs the closed-graph/section argument.
REFUTED notation/variance	Lemma A.3's displayed Mayer–Vietoris sequence as homology	It is cohomology. The stated groups survive after correction and UCT.

2 Referee determination

Recommendation: Major revision and resubmission after independent specialist replication. The complete manuscript should not be dismissed as arithmetically incoherent or as automatically refuted by the printed CDP20 conclusion. It also should not yet be represented as an established solution of the Hopf problem.

Revised central conclusion: No fatal internal contradiction has been identified in the complete 108-page manuscript. The direct algebraic-dimension theorem is now assessed as substantially validated, conditional on the earlier period-family construction. The nearby-cycle/Leray topology route is now assessed as a logically sufficient alternative to the global collapse for the sphere-recognition conclusion. These are meaningful upgrades from the 24 August review.

What remains unproved by this audit: Exact algebra is not the main uncertainty. The remaining high-risk obligations are the geometric identification of the based meridians and their drifts; the realization of the stated local specialization maps by the actual cusp and multiple-fibre models; the integral nearby-cycle edge map and its supported Gysin comparison; the first Leray transgression as the same integer p appearing in van Kampen; and the analytic duality/base-change/literature comparison in Section 10.

The manuscript claims a compact complex threefold X and a holomorphic map

$$f : X \longrightarrow \mathbb{P}^1$$

whose general fibres are complex two-tori, whose three special fibres consist of a reduced irreducible nonnormal normal-crossing surface W and two multiple fibres $3S_1$ and $4S_2$, and for the chosen logarithmic twisting data satisfies

$$\pi_1(X) = 1, \qquad H^*(X;\mathbb{Z}) \cong H^*(S^6;\mathbb{Z}), \qquad X \cong_{\mathrm{diff}} S^6.$$

It further claims

$$a(X) = 1, \qquad K_X \cong f^*\mathcal{O}_{\mathbb{P}^1}(-1) \otimes \mathcal{O}_X(2S_2),$$

$$R^1 f_*\mathcal{O}_X \cong \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2 f_*\mathcal{O}_X \cong \mathcal{O}(-1), \qquad c_3(X) = 2.$$

If correct, these claims give a complex structure on the standard smooth six-sphere and contradict printed conclusions of Campana, Demailly, and Peternell. The burden of proof is therefore exceptional.

The disposition has three levels.

1. **Finite and formal mathematics:** The displayed integer matrices, exterior powers, Smith forms, invariant and coinvariant lattices, period identities, Neron–Severi kernel, intersection form, Appendix B rank bookkeeping, and corrected Lemma A.3 map have been independently reproduced.

2. **Conditional proof reconstruction:** The complete direct algebraic-dimension chain and the formal nearby-cycle/Leray-to-homology chain survive line-by-line reconstruction under explicitly isolated geometric inputs.
3. **Publication-level certification:** The global construction, the bridge from local models to the topological and sheaf-theoretic maps, and the Section 10 analytic/literature claims still require independent domain specialists. Failure to locate an error is not a certificate of an extraordinary theorem.

3 Scope, source status, and review method

3.1 Source completeness and provenance

The reviewed archive, `s6_canonical_108_pages_audited.zip`, is complete as a reconstruction package. It contains ten modular source parts, a modular assembler, a monolithic source, a 108-page compiled PDF, an authority reassembly, build scripts, checksum manifests, page-by-page visual audits, and reconstruction provenance. The archive's own verification script passes. The modular and monolithic sources compile to byte-identical PDFs in the audited environment.

This establishes fidelity to the supplied authority PDF, not authorship of the TeX source and not mathematical correctness. The archive expressly states that it is an independent reconstruction rather than the author's original TeX. For any formal submission, an author-controlled source or an explicit author sign-off on the reconstructed canonical PDF is necessary.

The earlier review package was based on an incomplete 155-line assembler plus packet 06. Its conclusion that the main theorem was directly contradicted by CDP20 was a correct description of the logical conflict visible in that slice, but an incorrect final assessment of the complete manuscript because Section 10 was unavailable. This report replaces that review in full.

3.2 Review procedures

The audit used four complementary procedures.

First, I read the complete proof dependency chain rather than reviewing isolated claims. In particular, I traced the construction from the monodromy representation through the period family, the three local fillings, the global gluing, both integral topology computations, smooth recognition, analytic invariants, and the literature-conflict analysis.

Second, I independently reproduced exact computations with integer and symbolic arithmetic. The legacy harness contains twenty-eight passing checks. A second supplemental harness prompted by the Codex log contains fourteen passing checks, including the corrected modular-frame automorphy, the current Lemma 9.3 simplification, the cohomological Mayer–Vietoris matrix behind Lemma A.3, the two Appendix B lattice maps, the rank-forcing argument in B.5–B.6, and the common Leray transgression integer. These are forty-two check executions across two harnesses, with some intentional overlap; they are not advertised as forty-two logically independent theorems. The separate Codex claim of a 38/38 core harness is recorded but not verified because that harness was not supplied.

Third, I reconstructed the dependency graph of the two topology routes. The first uses an explicit collapse and Mayer–Vietoris. The second uses integral nearby cycles and the Leray spectral sequence. The second route does not depend on Proposition 7.2 for sphere recognition; its own indispensable bridges are now listed separately. Agreement remains significant, but the two routes are not treated as independent where they share local gluing, meridian, or specialization data.

Fourth, I reviewed Section 10 as an argument about exact proof mechanisms rather than as an appeal to exception. The review distinguishes the local conductor lemma, its global application to W through duality and base change, the precise comparison with the printed CDP reductions, and the logically later assertion that the published conclusion is falsified by the candidate.

3.3 Meaning of audit labels

"Verified computationally" means that the displayed finite calculation has been reproduced exactly. "No contradiction found" means that the proof was read and its logical transitions appear coherent, but it is not a certification. "Major revision request" identifies a pivotal argument whose present exposition is too compressed for an extraordinary theorem, even where no specific counterexample has been found. "Confirmed error" is reserved for a statement or proof sentence for which a definite defect can be exhibited.

4 The proof architecture and its dependency graph

The main theorem has the following dependency chain.

1. Sections 2–3 construct a rank-four integral local system over the $(3, 4, \infty)$ orbifold and a nonpolarized holomorphic family of complex two-tori with monodromies of orders 3, 4, and unipotent type at the cusp.
2. Section 4 constructs a toric filling at the cusp from an infinite A_2 fan and a free, properly discontinuous $\mathbb{Z}^2$ action. Its central fibre W is irreducible and nonnormal, with normalization dP_6, three double curves, and two triple points.
3. Section 5 constructs the order-3 and order-4 logarithmic fillings, with reduced fibres bielliptic surfaces.
4. Section 6 glues the smooth family and three fillings into a compact complex threefold $X \to \mathbb{P}^1$ and records the discrete gluing parameter ℓ_0.
5. Section 7 computes $\pi_1(X)$ and integral homology. For the chosen twists $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, the controlling integer
$$p = 12\ell_0 - 4\ell_1 - 3\ell_2$$
is -1. The manuscript gives two integral homology computations.
6. Section 8 uses the simply connected integral homology-sphere conclusion, Hurewicz–Whitehead theory, and $\Theta_6 = 0$ to identify X diffeomorphically with S^6.
7. Section 9 computes algebraic dimension, direct images, the canonical bundle, Chern-number consequences, holomorphic forms, and automorphisms.
8. Section 10 explains why the printed CDP98/CDP20 proof mechanisms are claimed not to cover this X.

A fatal defect in Sections 2–6 would invalidate the space. A defect in either topology route could potentially be repaired by the other only if their shared local specialization and gluing data remain valid. A defect in Section 9 would not by itself destroy the complex structure on S^6, but it could remove the claimed contradiction with CDP20. A defect in Section 10 would reinstate the literature obstruction unless the published theorem itself were independently corrected.

4.1 Conditional recognition theorem and obligation split

The minimal logical route to the smooth-sphere conclusion can be stated without the global collapse.

Conditional recognition theorem. Assume the following five inputs.

- **(G)** Sections 2–6 produce a closed connected smooth six-manifold X with a proper map $f : X \to \mathbb{P}^1$ and the stated smooth and special fibres.
- **(V1)** The based van Kampen inclusions, primitive meridians, signs, logarithmic drifts, and the cusp null-homotopy give the presentation of Theorem 7.17 with $p = 12\ell_0 - 4\ell_1 - 3\ell_2$.
- **(B1)** At the cusp, integral specialization is injective with image $H^q(F; \mathbb{Z})^{T_0}$ in every degree.
- **(F1)** At the two finite fibres, the specialization lattices and indices are those of Proposition 7.14 and Lemma A.4.
- **(L1)** The Leray transgression $d_2^{0,1}(12\gamma)$ is geometrically the obstruction class $\pm p\omega$, and the multiplicative identifications on the E_2 page are compatible with the stated invariant/coinvariant generators.

Then, for the actual data $p = -1$, the underlying smooth manifold of X is diffeomorphic to S^6.

Proof. Under (V1), $\pi_1(X) \cong \mathbb{Z}/|p| = 1$. Under (B1), (F1), and the already verified monodromy algebra, Proposition 7.26 gives the stated E_2 page. Under (L1), multiplicativity gives all three transgressions as multiplication by the same nonzero integer p. When $p = -1$, the relevant kernels and cokernels vanish, so $H^1(X) = H^2(X) = H^3(X) = 0$. Universal coefficients imply $H_1(X) = H_2(X) = 0$; Poincare duality then gives $H_3(X) = H_4(X) = H_5(X) = 0$, while $H_0(X) = H_6(X) = \mathbb{Z}$. Hence X is a simply connected integral homology six-sphere. Section 8's Hurewicz–Whitehead argument makes it a homotopy sphere, and $\Theta_6 = 0$ gives the diffeomorphism. □

The obligations attached to Proposition 7.2 should therefore be split from this minimal set:

- **(C1)** a finite equivariant CW description and global cusp-collapse retraction;
- **(C2)** the integral collapse matrix, including orientations, torsion, and saturation;
- **(C3)** equality of collapse specialization with nearby-cycle specialization;
- **(MV1)** geometric identification of the printed Mayer–Vietoris maps.

These remain necessary to certify the first topology route, the explicit collapse map, the stated generators of $H_*(W)$, and some later normalizations. They are not required for the conditional theorem above.

5 Section-by-section mathematical review

5.1 Lattice and monodromy data (Section 2)

The matrices

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

satisfy

$$T_1^3 = I, \qquad T_2^4 = I, \qquad T_0 = (T_1 T_2)^{-1} = I + N, \qquad N^2 = 0.$$

The exact orders are indeed 3 and 4. The displayed contragredients $A_j = (T_j^{-1})^t$ and $M_0 = (T_0^{-1})^t$ are correct, and $A_1 A_2 M_0 = I$. The invariant and vanishing-cycle lattices are also correct:

$$\ker(M_0 - I) = \operatorname{im}(M_0 - I) = \langle \widehat{w}, \widehat{\delta} \rangle,$$

and the induced map

$$B_0 : \overline{\Lambda} \to \Lambda_{\mathrm{tor}}, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

is unimodular. The fixed lattices

$$\Lambda^{A_1} = \langle \varepsilon, \widehat{\delta} \rangle, \qquad \Lambda^{A_2} = \langle \varepsilon', \widehat{\delta} \rangle$$

are primitive rank-two sublattices with the stated generators.

The invariant alternating form Q_0 is also consistent. In the ordered basis (γ, u, w, δ), its nonzero entries are $Q_0(\gamma, \delta) = 1$ and $Q_0(u, w) = 6$. Direct multiplication gives $T_j^t Q_0 T_j = Q_0$. The induced form

$$b_N(x, y) = Q_0(x, Ny)$$

on $V / \ker N$, in the basis $\bar{w}, \bar{\delta}$, is $\mathrm{diag}(6, -1)$. This confirms the manuscript's central nonpolarized feature: the monodromy-compatible alternating form has indefinite limiting form and does not define a polarization. This matters because standard polarized degeneration theorems cannot simply be imported (Clemens, 1977; Mumford, 1972).

The group-theoretic uniformization by the $(3, 4, \infty)$ triangle group is standard. The manuscript carefully separates the uniformizing coordinate from the period coordinate at the cusp and derives the translation sign. I found no inconsistency in the presentation $\Delta \cong \mathbb{Z}/3 * \mathbb{Z}/4$ or the definition of the representation.

Disposition: Exact algebra verified. No correction required, except that a concise table of all actions on V, Λ, and their exterior powers would reduce later sign risk.

5.2 Period family (Section 3)

The period matrix is built from

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta,$$

with $F^1 = \langle \sigma_1, \sigma_2 \rangle$. The construction proceeds through three nontrivial analytic steps: a modular function τ, a torsor solution μ for an automorphy sheaf of degree -1, and a solution β to a further additive cocycle. The final imaginary constant is chosen so that the lattice remains nondegenerate.

The modular transformation laws used later are mutually consistent. The automorphy sheaf calculation in Lemma 3.10, including the degree -1 conclusion, is coherent: the chosen modular expression has the required character, and its divisor behavior at the cusp gives $\mathcal{L} \simeq \mathcal{O}(-1)$. Because a square-root factor is used, the argument would be clearer if formulated invariantly in terms of the orbifold line bundle and its character, rather than by an implicitly chosen branch. I did not find a contradiction in the present construction.

The nondegeneracy argument is important. It is not enough that $\Pi(z)$ be formally equivariant; its four real columns must span a full lattice in $\mathbb{C}^2$. The manuscript reduces the determinant to an inequality involving

$$D = \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}$$

and uses the free imaginary constant in β to force $D < 0$ throughout a compact fundamental domain. The proper-discontinuity and compactness input makes this strategy plausible. The same expression later controls the signature of the Hermitian form associated with η.

The cusp asymptotic decomposition

$$Z = sB_0 + C(t_c)$$

is the precise bridge to the toric filling. I checked the equivariance and the exponential identity that produces the torus element $c_{\bar{\lambda}}(t_c)$. No algebraic inconsistency was found.

Disposition: No fatal defect found. For publication, the torsor existence and the global nondegeneracy choice should be stated as standalone propositions with explicit cohomology groups and norm estimates. The current proof is credible but densely packed.

5.3 Toric filling at the cusp (Section 4)

The fan consists of cones over the cells of the periodic A_2 triangulation at height one. Every maximal cone is unimodular because the edge differences form a $\mathbb{Z}^2$ basis. Local finiteness of the triangulation gives a Hausdorff, second-countable toric complex manifold, and the character $t = \chi^{(0,0,1)}$ has reduced normal-crossing divisor

$$t^{-1}(0) = \bigcup_{v \in \mathbb{Z}^2} E_v.$$

The action

$$\Psi_{\bar{\lambda}} = \tau_{\bar{\lambda}} \circ \Phi_{\bar{\lambda}}$$

combines integral translation of the fan with a t-dependent torus translation. The manuscript does more than assert properness: it introduces a rescaled position map and obtains a quantitative displacement estimate. Since B_0 is orthogonal and unimodular, large group elements move compact sets apart uniformly near the cusp. The compact fundamental-set argument then yields properness of the quotient map to the disc.

The action is free. On the central fibre, the fan translation is free on irreducible components; away from the central fibre, the action corresponds to a free lattice translation on the torus cover. The quotient therefore has local models

$$t = z_0, \qquad t = z_0 z_1, \qquad t = z_0 z_1 z_2,$$

and is smooth. The invariant logarithmic/top form gives the asserted relative canonical trivialization.

The normalization and conductor combinatorics are consistent. A fundamental component is dP_6; opposite sides of its toric hexagon are identified. This yields three rational double curves and two triple points. The orbit stratification

$$W = (\mathbb{C}^*)^2 \sqcup 3\mathbb{C}^* \sqcup \{P, Q\}$$

gives $e(W) = 2$.

I found one definite exposition defect. Proposition 4.7(iv) says the image of the section is closed because it is "proper over Δ." That is not the correct reason when the base is an open disc. The conclusion is readily repaired: a continuous section of a Hausdorff map has closed graph/image, or one may prove closedness directly in the chosen quotient chart. This does not threaten the construction.

The highest-risk part of Section 4 is not a visible algebraic error but the global topological bookkeeping for an infinite fan and quotient. The manuscript contains the needed ideas, yet a publication version should consolidate them into one theorem: local finiteness, second countability, freeness, proper discontinuity, compact fundamental domain, and compatibility of the toric deformation with the group action.

Disposition: No theorem-breaking defect found. One minor proof sentence should be corrected; global quotient topology deserves expansion.

5.4 Finite-order logarithmic fillings (Section 5)

The local family over the linearizing disc carries an affine transformation whose linear part is A_j and whose translation part is v_j/m_j. The admissibility conditions ensure that the affine cyclic action is free. The fixed-point criterion is correctly reduced to membership of the translation vector in $(A_j - I)$ of the torus lattice, and the chosen vectors avoid that image.

The quotient produces a smooth threefold with multiple fibre $m_j S_j$. The reductions S_j are bielliptic surfaces. The computed first homology and torsion are compatible with the standard Bagnera–de Franchis classification (Barth et al., 2004). The order-3 and order-4 cases also generate the precise local specialization indices later used in Section 7.

The logarithmic regluing section is arranged so that the transformed zero section is invariant. The conjugation identity used to compare the punctured family with the quotient is correct. The resulting drift controls $\ell_j = \gamma(v_j)$.

Disposition: No fatal error found. The proof would be more auditable with a single diagram showing the cover, affine action, quotient, multiple fibre, reduction, and boundary identification. The classification of the two bielliptic surfaces should be tied explicitly to the standard notation in Barth et al. (2004).

5.5 Global gluing and compactness (Section 6)

The three local models are glued to the smooth family over B°. The overlap maps are biholomorphisms by construction. The manuscript addresses Hausdorffness through shrinkable collars and the fact that each local piece maps properly to a disc. Properness of the global map is local on the base for a separated holomorphic map; the resulting total space is compact because the base is compact and the map is proper.

The zero section is used as more than notation: it fixes the lift of boundary meridians and prevents hidden fibre translations in the finite-order gluings. At the cusp, affine regluing changes the discrete parameter ℓ_0, and the tautological gluing gives $\ell_0 = 0$.

No incompatibility was found among the boundary identifications. The proof is nevertheless unusually sensitive to the distinction between the smooth fibre lattice, the toric translation lattice, and the quotient of Λ by Λ_{tor}. A one-page gluing table should identify, for every boundary component, the deck group, meridian, fibre translation, orientation, and induced relation in π_1.

Disposition: Coherent. Major revision only in presentation and auditability, not because of a demonstrated flaw.

5.6 Fundamental group (Section 7.5)

After attaching the cusp filling, the normal closure of Λ_{tor} is $\ker\gamma$. Hence the image of the fibre group is generated by one central element c. The two finite fillings impose

$$x^3 = c^{\ell_1}, \qquad y^4 = c^{\ell_2},$$

and the cusp relation gives $xy = c^{\ell_0}$. Thus

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \rangle.$$

Eliminating y makes the group abelian with relation matrix

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -(12\ell_0 - 4\ell_1 - 3\ell_2).$$

Because the entries include 3 and 4, the first Smith invariant is 1; hence

$$\pi_1(X) \cong \mathbb{Z}/|p|, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

The admissibility congruences imply $\gcd(p, 12) = 1$. For $(0, 1, -1)$, $p = -1$ and the group is trivial. For the comparison sign, $p = -7$.

These calculations pass exact Smith-normal-form checks. The manuscript's long sign lemma is not redundant: a relative sign error would interchange the simply connected and $\mathbb{Z}/7$ examples. I found its treatment internally consistent. The final invariant $|p|$ is orientation-independent, but the manuscript should provide a compact convention table because the proof currently requires the reader to track clockwise meridians through several covers.

Disposition: Algebra verified. No mathematical correction required.

5.7 First integral homology route: collapse and Mayer–Vietoris (Sections 7.1–7.6)

The first route constructs an equivariant collapse of a nearby torus fibre onto the singular fibre and uses Mayer–Vietoris to obtain

$$H_k(W;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}), \qquad H_k(\mathcal{M};\mathbb{Z}) \text{ of ranks } (1, 3, 6, 6, 3, 1),$$

followed by the cyclic groups $\mathbb{Z}/|p|$ in degrees 1 through 4 for X. The exterior-power matrices, saturated image lattices, rank of α_2, and primitive left kernel all pass exact checking.

The Codex-directed reinspection changes two aspects of this section.

First, Proposition 7.2 is no longer classified as a bottleneck for sphere recognition because Section 7.7 and Appendix B bypass it. It remains a high-risk theorem for the manuscript's explicit collapse description and first homology route. The appropriate obligations are (C1)–(C3) and (MV1), not an undifferentiated request to "check the topology."

Second, Lemma A.3 has a confirmed variance error. The displayed sequence

$$\cdots \to H_k(W) \to H_k(\widetilde{W}) \oplus H_k(D) \to H_k(\widetilde{D}) \to H_{k+1}(W) \to \cdots$$

has the direction of the **cohomological** Mayer–Vietoris sequence. The subsequent map is restriction

$$H^2(\widetilde{W}) \oplus H^2(D) \longrightarrow H^2(\widetilde{D}),$$

computed by intersection numbers on dP_6 and pullback from the three double curves. Its integral matrix has rank 4 and Smith diagonal $(1, 1, 1, 1)$; hence kernel rank 3 and free cokernel rank 2. The preceding connecting homomorphism $H^1(\widetilde{D}) \to H^2(W)$ contributes one additional primitive $\mathbb{Z}$. Thus

$$H^\bullet(W;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}),$$

all free, and universal coefficients recovers the same homology groups. The lemma's conclusion survives; its written proof must be corrected.

Disposition: Exact finite algebra passes. The route is conditionally coherent after the Lemma A.3 variance repair. Its global collapse and map-identification claims remain specialist-verification items, but they are not part of the minimal recognition theorem.

5.8 Second integral homology route: Leray spectral sequence (Section 7.7)

This route is now assessed as an independent minimal topology engine rather than merely as corroboration. It depends on the actual geometric inputs (B1), (F1), and (L1), but it does not depend on the global collapse of Section 7.2.

Let $j : B^\circ \hookrightarrow B$ and let $\mathcal{L}^q$ have stalk $\bigwedge^q V$. The cusp theorem identifies $R^q f_*\mathbb{Z}$ with $j_*\mathcal{L}^q$ near p_0. The finite fibres contribute the quotient lattices

$$Q^1 = \mathbb{Z}/3 \oplus \mathbb{Z}/4, \qquad Q^2 = Q^3 = \mathbb{Z}/2 \text{ at } p_2, \qquad Q^4 = \mathbb{Z}/3 \oplus \mathbb{Z}/4.$$

The resulting global sections and coinvariants are

$$\begin{aligned} H^0(B, R^1) &= 12\mathbb{Z}\gamma, & H^2(B, R^1) &= \mathbb{Z}[\bar{\delta}], \\ H^0(B, R^2) &= \mathbb{Z} \cdot 2q, & H^2(B, R^2) &= \mathbb{Z}[\gamma\delta], \\ H^0(B, R^3) &= \mathbb{Z} \cdot 2\gamma uw, & H^2(B, R^3) &= \mathbb{Z}[uw\delta]. \end{aligned}$$

The parabolic groups $H^1(B, R^1)$ and $H^1(B, R^2)$ vanish by explicit lattice-intersection calculations. These intersections and saturation statements have been reproduced exactly.

The first transgression is not merely a symbolic parameter. It is computed by choosing lifts over the three special discs and over a pair-of-pants neighbourhood, evaluating their discrepancies on clockwise meridians, and using

$$\widetilde{a}_0 = t_\mu \hat{\rho}_2^{-1} \hat{\rho}_1^{-1}.$$

The sum of the three discrepancy values is $-p$, giving

$$d_2^{0,1}(12\gamma) = \pm p\omega.$$

The remaining transgressions are determined by the Leibniz rule. If

$$d_2(2q) = p'\omega\bar{\delta}, \qquad d_2(2\gamma uw) = p''\omega[\gamma\delta],$$

then the relation $(12\gamma)(2q) = 12(2\gamma uw)$ gives

$$p'' = 2p - p'.$$

The vanishing of $(2q)(2\gamma uw)$ and the two primitive pairings give $|p'| = |p''|$. Since $p \neq 0$ under admissibility,

$$(p')^2 - (p'')^2 = 4p(p' - p) = 0$$

forces $p' = p'' = p$. This exact argument passes the supplemental harness.

For $p = -1$ the three maps are units. The spectral-sequence deductions, universal-coefficient step, and Poincare-duality step are correct. The unresolved question is therefore sharply geometric: do the actual local models realize

the specialization lattices and based discrepancy class exactly as asserted? This is the combined content of (V1), (B1), (F1), and (L1).

Disposition: Conditionally validated as a complete topology route. Independent specialist replication should target the geometric bridges, not repeat the already verified Smith and spectral-sequence arithmetic.

5.9 Recognition as the smooth six-sphere (Section 8)

Once X is simply connected and has the integral homology of S^6, the Hurewicz theorem and Whitehead's theorem make it a homotopy 6-sphere. Smooth homotopy 6-spheres are classified by Θ_6, and $\Theta_6 = 0$ (Kervaire & Milnor, 1963). Thus the diffeomorphism conclusion follows.

This step is standard and correct, conditional on Section 7. The manuscript could make explicit that the smooth structure is oriented by the complex orientation, although in dimension six there is no exotic sphere ambiguity.

Disposition: Valid conditional deduction.

5.10 Algebraic dimension (Section 9.1)

This section is upgraded from "no contradiction found" to **conditionally validated**. The proof is complete in the audited manuscript, and it is independent of Section 7 and of CDP20.

For $E \in H^2(F_b;\mathbb{Z})$, Lemma 9.2 proves that E is of type $(1,1)$ precisely when

$$E \wedge \sigma_1 \wedge \sigma_2 = 0,$$

which is the equation $P_E(z) = 0$ with

$$P_E = E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta + E_{w\delta}(6\mu^2 - \tau\beta).$$

The wedge expansion and all signs have been reproduced. Lemma 9.3 uses the g_0 and g_2 transformation laws to show that

$$1, \tau, \mu, \beta, 6\mu^2 - \tau\beta$$

are linearly independent. The two subtraction identities, including the corrected algebraic simplification reported by Codex, are correct in the current source. Consequently

$$P_E \equiv 0 \quad \Longleftrightarrow \quad E \in \mathbb{Z}\eta, \qquad \eta = uw + 6\gamma\delta.$$

For every smooth fibre, η is a $(1,1)$ class with $\eta^2 = 12$. The Hermitian form has determinant

$$\frac{2}{3}\operatorname{Im}\tau\left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}\right) < 0,$$

so every nonzero multiple is nondegenerate and indefinite. Outside a countable set, $\mathrm{NS}(F_b) = \mathbb{Z}\eta$. The complex-torus algebraic-reduction theorem then gives a direct dichotomy:

- algebraic dimension two would supply an ample class with positive-definite Hermitian form;
- algebraic dimension one would supply the nonzero pullback of a point from an elliptic quotient, whose square is zero.

Both contradict the properties of $n\eta$. Therefore the very general fibre has algebraic dimension zero.

Lemma 9.6 supplies the global descent. If a meromorphic function is constant on uncountably many fibres, the analytic set

$$Z = \{dg \wedge f^*dt = 0\}$$

must be all of the complement of its polar divisor over B°; otherwise its countably many irreducible components could contain only countably many distinct fibre components. The function is therefore fibrewise constant and descends holomorphically away from finitely many base points. A transverse analytic disc gives polynomial growth and hence meromorphic extension across each missing point. Thus

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction.

The remaining conditions are upstream rather than internal to this proof: the period functions must define the claimed family and satisfy the nondegeneracy inequalities, and the standard structure theorem for algebraic reduction of complex tori must be cited in its exact form. No Section 7 topology and no CDP theorem is used.

Disposition: Conditionally validated; revise the manuscript mainly by making the standard torus theorem and the connectivity/irreducibility facts in the descent lemma citation-explicit.

5.11 Canonical bundle and direct images (Sections 9.2–9.4)

The relative two-form $e = d\zeta_1 \wedge d\zeta_2$ transforms by $\det R_g$. Its behavior at the finite orbifold points gives fractional vanishing orders $2/3$ and $1/4$, and it is regular and nonvanishing at the cusp. The resulting direct image is

$$f_*\omega_{X/B} \cong \mathcal{O}_{\mathbb{P}^1}(1).$$

The multiple-fibre correction yields

$$K_X \cong f^*\mathcal{O}(-1) \otimes \mathcal{O}(2S_2).$$

The character arithmetic is coherent. The non-torsion conclusion follows because the stated bundle has a nontrivial multiple-fibre component and its appropriate power is a nontrivial pullback.

For the structure-sheaf direct images, the manuscript obtains

$$f_*\mathcal{O}_X = \mathcal{O}, \qquad R^1f_*\mathcal{O}_X \cong \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2f_*\mathcal{O}_X \cong \mathcal{O}(-1).$$

The fibre cohomology at W is computed by the normalization/conductor sequence. The finite multiple fibres are handled by equivariant local models and descending induction through the nilpotent thickening. Relative duality identifies the top direct image. The eigenweights determine the splitting of R^1.

I found no algebraic inconsistency, and the resulting global Hodge numbers agree with the Leray calculation. The main concern is the precision of the analytic foundations. In a theorem of this importance, references to Grauert, base change, and relative duality should be attached to exact statements and hypotheses. The reader should not have to infer that the scheme-theoretic multiple fibres are covered, that the relevant sheaves are flat, and that duality applies to the Gorenstein nonnormal surface W.

Disposition: Plausible and internally consistent. Major revision requested for exact analytic theorem statements.

5.12 Holomorphic forms, automorphisms, and Chern data (Sections 9.5–9.7)

The vanishing of global holomorphic p-forms is derived by writing invariant forms on the universal cover and using the period derivatives and local growth at the cusp and orbifold points. The weight-zero and weight-one lemmas eliminate the remaining coefficients. The argument is technically detailed and consistent with the computed $h^{0,q}$.

The identity component of the automorphism group must preserve the algebraic reduction and hence acts fibrewise. Translation vector fields correspond to the invariant lattice line $\Lambda^\Delta = \mathbb{Z}\widehat{\delta}$, giving a one-dimensional vertical field. Its flow is $\mathbb{C}^*$ rather than an elliptic translation group because of the degeneration. The toric fixed set is one rational double curve with normal weights $(+1, -1)$. This is compatible with $e(X^{\mathbb{C}^*}) = e(X) = 2$.

Because $H^2(X;\mathbb{Z}) = H^4(X;\mathbb{Z}) = 0$, c_1 and c_2 vanish integrally. The top Chern number equals the Euler characteristic,

$$\langle c_3(T_X), [X] \rangle = e(X) = 2.$$

The Riemann–Roch arithmetic giving $\chi(\mathcal{O}_X) = 0$, $\chi(\Omega^1_X) = -1$, and $\chi(T_X) = 1$ is consistent.

Disposition: No contradiction found. These conclusions remain conditional on the topology and direct-image calculations.

6 Appendix B: integral specialization at the nonnormal cusp

Appendix B is the central bridge in the collapse-free topology proof. The Codex log correctly identifies it as the place where exact matrices alone are insufficient. A line-by-line reconstruction nevertheless supports a stronger disposition than the previous report: the formal nearby-cycle, hypercohomology, rank, saturation, and supported-Gysin chain is coherent and complete once the stated local toric models and standard integral nearby-cycle functoriality are granted.

6.1 Nearby-cycle sheaves and their integral gluing

The local equations $t = z_0$, $z_0 z_1$, and $z_0 z_1 z_2$ give Milnor fibres of homotopy type point, S^1, and T^2. Hence

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \qquad \mathcal{H}^1\psi = \mathcal{V}, \qquad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q.$$

The normalization globally distinguishes the two branches along each double curve. With the manuscript's orientation convention, the three restrictions at both triple points satisfy

$$n_1 - n_2 + n_3 = 0.$$

This gives

$$0 \to \mathcal{V} \to \bigoplus_{k=1}^{3} \iota_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \to 0, \qquad \delta_1 = \begin{pmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

The matrix has rank one and primitive image. The claimed cohomology

$$H^\bullet(W; \mathcal{V}) = (\mathbb{Z}^2, \mathbb{Z}, \mathbb{Z}^3, 0)$$

follows integrally.

6.2 The bottom row and the Lemma A.3 repair

The normalization/conductor resolution of $\mathbb{Z}_W$ yields the map

$$d : H^2(\mathrm{dP}_6; \mathbb{Z}) \to \mathbb{Z}^3, \qquad d(aH - \textstyle\sum b_i E_i) = (s, -s, s), \quad s = b_1 + b_2 + b_3 - a.$$

Its image is primitive. Thus

$$H^\bullet(W; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}),$$

all free. This is the cohomological calculation that Lemma A.3 writes with the wrong variance. Appendix B itself uses the correct cohomology direction.

6.3 Rank forcing and the degree-three supported class

The E_2 ranks are

$b=2$	2	0	0	0	0
$b=1$	2	1	3	0	0
$b=0$	1	2	4	2	1
	$a=0$	$a=1$	$a=2$	$a=3$	$a=4$.

Writing the five possible differential ranks as $(\alpha, \beta, \gamma, \delta, \epsilon)$, total-degree ranks force

$$\alpha = \gamma = 0, \qquad \beta + \delta + \epsilon = 1.$$

The supported argument takes a compact T_f-orbit $K \cong T^2$ in W°, transports it smoothly to a translate K_t of the saturated linear subtorus in F_t, and compares

$$H^3(W, W \setminus K) \to H^3(W) \xrightarrow{\mathrm{sp}^3} H^3(F_t)$$

with the Gysin map

$$H^1(K_t) \xrightarrow{\iota_!} H^3(F_t).$$

Because $F_t \cong T_f \times T_B$ topologically, $\iota_!(a) = a \times [T_B]^\vee$ is injective with saturated image of rank two. Hence sp^3 has rank two, forcing

$$(\alpha, \beta, \gamma, \delta, \epsilon) = (0, 0, 0, 1, 0).$$

The remaining filtration quotients are free, so the specialization images are saturated. Their ranks equal the ranks of the T_0-invariants, and inclusion into the invariant lattice therefore becomes equality.

The supplemental harness reproduces the two integral matrices and the complete rank-forcing logic.

6.4 Exact residual obligations

The formal proof is sound under the following items, which should be citation- and convention-complete in a revision.

1. The local quotient charts of Theorem 4.5 must really be holomorphically equivalent to the three stated normal-crossing equations, with the stated branch labels and compact-torus action.
2. The specialization morphism $\mathbb{Z}_W \to \psi_f \mathbb{Z}$ and its hypercohomology edge map must agree with the sheaf specialization used in Section 7.7 over the integers.
3. The smooth supported transport in Lemma B.3 must be tied functorially to that edge map. The map Θ is a smooth open embedding, which is sufficient for singular cohomology; it should not be read as holomorphic.
4. The marking must identify the transported T_f-orbit with the saturated sublattice Λ_{tor} and hence with the primitive Kunneth summand used by the Gysin map.
5. The first Leray transgression must be compared with the based meridian/gluing data without hidden sign or translation changes.

These are precise bridge lemmas, not missing matrix computations.

Disposition: Conditionally validated as an integral specialization proof, with specialist replication required for the five geometric/functorial inputs above.

7 Section 10 and the conflict with CDP98/CDP20

7.1 Four distinct propositions that must not be conflated

The literature issue decomposes into four propositions.

1. **Local proposition:** a reduced nonnormal normal-crossing fibre can carry a conductor-supported ambient cotangent section invisible in the torsion-free intrinsic cotangent sheaf.
2. **Example-specific analytic proposition:** the line bundles arising from X satisfy the normalization, duality, and top-degree base-change hypotheses needed to turn that section into $R^2 f_*(T_X \otimes L) \neq 0$.
3. **Proof-comparison proposition:** the printed CDP reduction uses an implication invalidated by this local phenomenon, or otherwise fails to account for it.
4. **Counterexample proposition:** the fully constructed X satisfies every hypothesis of the printed theorem and violates its conclusion.

The first proposition is independently reconstructed and strong. The second appears plausible and internally coherent but requires exact analytic references. The third requires specialist comparison with the original proof. The fourth additionally depends on certification of the entire construction and topology. A possible defect in a proof, falsity of a theorem, and correctness of the candidate are logically separate.

7.2 The formal contradiction

If the manuscript's main conclusions hold, then X satisfies

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1,$$

and has a holomorphic algebraic reduction $f : X \to \mathbb{P}^1$. Campana et al. (2020, Theorem 2.2) print, under those hypotheses, $c_3(X) \leq 0$, and their Corollary 2.3 prints $a(X) = 0$ for a compact complex threefold homeomorphic to S^6. The manuscript instead has $c_3(X) = 2$ and $a(X) = 1$. Both sets of conclusions cannot simultaneously be correct.

The complete manuscript does not evade this issue. It isolates two exact proof mechanisms and argues that one is irreparably false at the nonnormal fibre W, while the other is a monodromy error that can be repaired for this example.

7.3 The nonnormal-fibre cotangent mechanism

Let S be a reduced normal-crossing fibre in a smooth threefold $\mathcal{X} \to \Delta$, with normalization $\eta : \widetilde{S} \to S$ and conductor D. Locally the defining function is $g = ux$, uxy, or $uxyz$. The differential $dg|_S$ is nonzero away from the conductor but restricts to zero on the conductor.

For a line bundle A on S whose pullback to $\widetilde{S}$ is trivial, choose a nowhere-zero pullback section e. The normalization/conductor Mayer–Vietoris sequence lets the pair

$$\big(\eta^*(dg|_S) \otimes e, 0\big)$$

descend to a nonzero section

$$s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A).$$

Its image in $\Omega^1_S \otimes A$ is supported on the conductor and is torsion. Therefore its image in the torsion-free quotient

$$\widetilde{\Omega}^1_S = \Omega^1_S / \mathcal{T}(\Omega^1_S)$$

is zero. If A is nontrivial, s does not arise from the conormal line. This produces exactly the type of ambient cotangent section that is invisible to a vanishing statement for $\widetilde{\Omega}^1_S \otimes A$.

This is the strongest part of the manuscript's rebuttal. The local formulas are elementary and the descent through the conductor sequence is explicit. The mechanism shows that the implication

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) = 0 \quad \Longrightarrow \quad H^0(S, \Omega^1_X|_S \otimes A) = 0$$

is not valid for nonnormal S merely from vanishing of the conormal contribution. For normal S, extension across codimension two can restore the argument; for the nonnormal surface W, the conductor supports the obstruction.

For the constructed W, every line bundle from X pulls back trivially to the normalization dP_6 because $c_1 = 0$ and $\mathrm{Pic}(\mathrm{dP}_6) \hookrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$. Serre–Grothendieck duality then gives

$$H^2\big(W, (T_X \otimes L)|_W\big)^* \cong H^0\big(W, \Omega^1_X|_W \otimes (L^* \otimes \omega_X)|_W\big) \neq 0.$$

Top-degree cohomology and base change imply $R^2 f_*(T_X \otimes L) \neq 0$. The manuscript therefore claims that the hypothesis used by Campana et al. (2020, Proposition 2.4) holds for no L on this X.

I found no local algebra error in this mechanism. The exact analytic statement required is the Serre–Grothendieck duality isomorphism for the Cartier, hence Gorenstein, compact surface W:

$$H^2(W, E)^\vee \simeq \mathrm{Hom}_W(E, \omega_W) \simeq H^0(W, E^\vee \otimes \omega_W),$$

with $E = (T_X \otimes L)|_W$ and $\omega_W \simeq \omega_X|_W$ by adjunction for the principal reduced fibre. The required base-change statement is the top-degree map

$$(R^2 f_*(T_X \otimes L)) \otimes k(c) \longrightarrow H^2(f^*(c), (T_X \otimes L)|_{f^*(c)}),$$

for every scheme-theoretic fibre, including the multiple fibres. The top-degree character makes the claim plausible, but the revision should cite a theorem whose hypotheses explicitly cover the proper flat analytic map and locally free sheaf used here. Because the application would invalidate a published theorem's conclusion for this example, this chain must be independently refereed by experts in nonnormal complex spaces, analytic duality, and cohomology and base change.

7.4 The monodromy-coinvariant correction

The second issue concerns a split group extension

$$1 \to K \to G \to Q \to 1.$$

The correct abelianization is

$$G^{\mathrm{ab}} \cong Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q,$$

where the second summand consists of coinvariants under conjugation. It is not generally $Q^{\mathrm{ab}} \oplus K^{\mathrm{ab}}$ unless the action is trivial. This is standard group homology.

For the manuscript's torus fibration, the unipotent monodromy acts nontrivially on $H_1(F; \mathbb{Z})$. The coinvariant rank is one, not four. Consequently, formulas in the printed CDP98/CDP20 route that use $b_1(F)$ rather than the coinvariant rank do not apply to this example. This also explains how

$$H^1(X, \mathcal{O}_X) \cong \mathbb{C}$$

can coexist with

$$H^1(F, \mathcal{O}_F) \cong \mathbb{C}^2$$

without surjectivity of restriction.

The manuscript then proves by a separate countability argument that a line bundle non-torsion on every fibre nevertheless exists. Thus the monodromy problem breaks the printed route to that input but can be repaired for this example. By contrast, the nonnormal cotangent reduction remains unavailable.

The group-theoretic correction is correct. Its application to the exact sequence in CDP98 should still be checked against the precise local topological model and notation in the original paper, but the mathematical principle is not in doubt.

7.5 Referee assessment of the literature dispute

The earlier packet-only report treated CDP20 as an external theorem that immediately refuted the manuscript. That treatment is no longer defensible after reading Section 10. The manuscript supplies a concrete countermechanism, not a generic assertion of exception.

The appropriate present conclusion is narrower:

- The manuscript and the printed CDP20 conclusions are formally incompatible.
- Section 10 gives a coherent explanation of why the printed proof does not cover the proposed nonnormal fibre.
- This review found no internal defect in that explanation.
- The explanation has not, by virtue of this review, been externally adjudicated. Acceptance would effectively publish a counterexample to CDP20 and should involve independent referees familiar with the original proof.

8 Confirmed errors, repairs, and reported prior corrections

8.1 Remark 7.28: quantifier error

The statement

$$|p| = 1 \quad \Longleftrightarrow \quad (\ell_1 \bmod 3, \ell_2 \bmod 4) \in \{(1,3),(2,1)\}$$

is false as a biconditional for a fixed triple (ℓ_0, ℓ_1, ℓ_2). For example,

$$(\ell_0, \ell_1, \ell_2) = (100, 1, -1)$$

has the listed residue pair but $p = 1199$. The correct statement is existential in ℓ_0: for fixed admissible (ℓ_1, ℓ_2), there exists an integer ℓ_0 with $|p| = 1$ exactly for those two residue pairs.

8.2 Proposition 4.7(iv): closed-image justification

The section image is not shown closed by calling the section "proper over the open disc." A section of a continuous map into a Hausdorff space has closed graph, and here $f_0 \circ \varsigma = \mathrm{id}$ gives the required closed-embedding argument directly. The conclusion is correct; the stated justification should be replaced.

8.3 Lemma A.3: variance error

The displayed Mayer–Vietoris sequence and the maps used in the proof are cohomological. Replace every H_k in that calculation by H^k, retain the rank and saturation computation, and then invoke universal coefficients to obtain the asserted homology. The supplemental exact matrix check confirms rank 4, primitive image, kernel rank 3, and free cokernel rank 2.

8.4 Corrections reported by Codex and present in the audited source

The current source uses the invariant frame $\mathsf{e} = F^{-1}e$ on source page 71. Since both F and e transform by $\jmath_g^{-1}$, the inverse factor is necessary and the current expression is invariant. The current version of Lemma 9.3 also has the correct second-difference identity. These are not current-source errors.

The cusp deck group is written as $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$, of rank two, while Λ remains the rank-four homology lattice of a smooth fibre. That typing is consistent in the audited passages inspected. Because the Codex log does not supply the location of an earlier $\Lambda/\bar{\Lambda}$ mistake, the review records the repair without inventing an additional confirmed defect.

9 Major revision requirements

9.1 1. Publish a typed bridge-lemma ledger

Every central claim should be tagged as exact algebra, standard theorem, geometric identification, analytic comparison, or literature comparison. The minimum topology set is (G), (V1), (B1), (F1), and (L1). Each should have a standalone statement, hypotheses, proof location, orientation conventions, and a falsification test.

9.2 2. Separate the collapse route from the minimal recognition route

Proposition 7.2 and the first Mayer–Vietoris route should be developed under (C1)–(C3) and (MV1). The paper should state prominently that the sphere-recognition theorem can instead use Section 7.7 and Appendix B. Conversely, claims about the explicit collapse map, named generators of $H_2(W)$, and equality with nearby-cycle specialization must not be inferred from the Leray route without proof.

9.3 3. Make the nearby-cycle and transgression bridges citation-complete

State the precise integral nearby-cycle functor, the specialization morphism, the edge-map convention, the supported transport map, and the Gysin orientation. Then give a commuting diagram comparing the first Leray transgression with the based meridian obstruction. Exact matrix agreement is not a substitute for this diagram.

9.4 4. Make analytic duality and top-degree base change theorem-complete

For Proposition 9.13 and Theorem 10.5, state the morphism, sheaf, flatness, fibre dimension, Gorenstein or Cohen–Macaulay hypothesis, and the exact base-change map. Treat the reduced nonnormal fibre and the nonreduced multiple fibres separately. A stable citation must cover the analytic category actually used.

9.5 5. Present the CDP comparison as an independently refereed note

Separate the local conductor proposition, the example-specific duality/base-change application, the monodromy-coinvariant correction, and the comparison with exact numbered statements in CDP98/CDP20. A proof gap should not be rhetorically converted into theorem falsity before the candidate is certified.

9.6 6. Add a complete convention and gluing table

List the lattice, quotient lattice, monodromy action, deck group, based meridian, orientation, affine translation, logarithmic drift, and induced relation at each special point. This table is the principal defense against a hidden sign or lattice-typing error.

9.7 7. Correct all three confirmed local defects

The revision ledger must explicitly correct Proposition 4.7(iv), Remark 7.28, and Lemma A.3. It should also identify the version in which the page-71 modular factor, Lemma 9.3 simplification, and $\Lambda/\bar{\Lambda}$ notation were repaired.

9.8 8. Strengthen source and proof provenance

The reviewed archive is a reconstruction. A formal submission needs author-controlled source or author sign-off on the canonical PDF. Distinguish "candidate proof," "independent repair or alternative," and "jointly surviving version" at the theorem level. The supplied Codex log is useful provenance, but unsupplied files and harnesses cannot be treated as independently audited evidence.

10 Computational audit

Two executable harnesses accompany this edition.

The legacy harness performs twenty-eight exact checks, all passing. It covers monodromy orders and contragredients, invariant and coinvariant lattices, Smith normal forms, Wang ranks, saturation of exterior-power images, the

fundamental-group relation matrix, period transformations, the Neron–Severi polynomial, η^2, and the Hermitian Gram determinant.

The supplemental Codex-integration harness performs fourteen checks, all passing:

- invariance of the corrected page-71 frame $F^{-1}e$;
- the current g_0 difference and corrected second-difference algebra in Lemma 9.3;
- the exact kernel $P_E \equiv 0 \Longleftrightarrow E \in \mathbb{Z}\eta$;
- non-isotropy of every nonzero multiple of η;
- the corrected cohomological Mayer–Vietoris matrix in Lemma A.3, with Smith diagonal $(1, 1, 1, 1)$;
- the primitive normalization-difference map in Appendix B;
- the primitive δ_1 map defining $\mathcal{V}$;
- all rank candidates in B.5 and the fact that the supported rank-two class forces $\delta = 1$;
- the factorization that forces $p' = p'' = p$ in Proposition 7.27;
- unit transgressions for the actual value $p = -1$;
- the fixed-triple counterexample to Remark 7.28;
- the rank consistency of $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$ as the cusp deck group.

There are therefore forty-two passing check executions across the two harnesses. Several deliberately overlap the same formulas under different logical decompositions. The number should not be read as a measure of theorem probability.

The Codex log separately reports a 38/38 "independent core harness." That code and its output were not supplied, so this review neither adopts nor contradicts the number. It records only independently rerunnable artifacts in the present ZIP.

No harness proves that a displayed matrix is the actual map induced by the geometric construction. Every computational result in this package is explicitly conditional on that identification.

11 Final recommendation

I recommend **major revision and resubmission**, followed by independent specialist replication.

The evidential balance is now more favorable than in either the packet-only review or the first full-manuscript review. Two major subarguments have survived a more exact reconstruction:

1. Section 9.1 gives a direct and logically independent proof of $a(X) = 1$ and $\mathcal{M}(X) = f^*\mathbb{C}(t)$.
2. Section 7.7 with Appendix B gives a collapse-free route from the local specialization and transgression data to the integral homology sphere.

The manuscript therefore should not be rejected on the grounds that algebraic reduction remains open, that Proposition 7.2 is the only topology engine, or that CDP20 can be cited without engaging Section 10.

Acceptance would still be premature. The candidate is extraordinary, and the unresolved tasks lie precisely where formal algebra offers the least protection: construction of the global analytic space, based gluing and meridian data, integral nearby-cycle functoriality in the actual toric quotient, geometric identification of Leray transgressions, and the analytic and bibliographic adjudication of the CDP conflict. The three confirmed local errors are repairable and do not presently destroy the main chain, but they show why a typed morphism and variance ledger is necessary.

The appropriate publication path is coordinated review by specialists in: (1) toric or semi-abelian degenerations and infinite locally finite fans; (2) integral nearby cycles, supported specialization, and topology of nonnormal normal-crossing degenerations; and (3) compact complex threefolds, analytic duality/base change, and the CDP98/CDP20 arguments. At least one referee should reconstruct the minimal recognition theorem from (G), (V1), (B1), (F1), and (L1) without relying on the manuscript's prose summary.

The present verdict is therefore neither certification nor dismissal. It is that the manuscript now has a coherent, nontrivial, and partially independently validated proof architecture worthy of specialist replication, with a sharply delimited list of remaining obligations.

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13 Appendix A: audit disposition by central claim

Central claim	Audit disposition	Principal dependency	Required follow-up
Smooth compact complex threefold X exists	No fatal defect found; not certified	Sections 2–6, obligation (G)	Independent review of period family, infinite toric quotient, finite fillings, and gluing
$\pi_1(X) = 1$	Exact group algebra verified, conditional on based attaching relations	(V1), Theorem 7.17	Based meridian, primitive generator, sign, drift, and cusp null-homotopy audit
$H^*(X;\mathbb{Z}) = H^*(S^6;\mathbb{Z})$	Leray deduction conditionally validated	(B1), (F1), (L1)	Replicate integral specialization and transgression comparison

Central claim	Audit disposition	Principal dependency	Required follow-up
Explicit collapse and first MV route	Algebra verified; global geometry open	(C1)–(C3), (MV1)	Cell-level equivariant construction and map ledger; repair Lemma A.3 variance
$X \cong_{\mathrm{diff}} S^6$	Standard conditional consequence	(G), (V1), (B1), (F1), (L1), $\Theta_6 = 0$	None beyond topology inputs
$a(X) = 1$ and f algebraic reduction	Conditionally validated by direct period/descent proof	Sections 2–3 and 9.1	Exact citation for complex-torus algebraic reduction; upstream period-family validation
Direct images and K_X	Plausible, not fully certified	Sections 9.2–9.4	Exact analytic base-change, relative duality, and multiple-fibre hypotheses
$c_3(X) = 2$	Conditional on homology-sphere topology	Sections 7–8	No separate arithmetic issue
Local nonnormal cotangent mechanism	Independently reconstructed	Lemmas 10.2–10.3	Present as standalone proposition
CDP20 conclusion fails for X	Substantive but externally unadjudicated	Entire main theorem plus Theorem 10.5 and literature comparison	Specialist review of duality, base change, and exact published proof

14 Appendix B: supersession and version history

Three review stages should remain visible.

1. The packet-06 report correctly identified the formal incompatibility with the printed CDP20 conclusion but lacked Section 10 and the full construction. Its rejection recommendation is withdrawn.
2. The full-manuscript report of 24 August 2026 withdrew that rejection and recommended major revision. Its principal overstatement was to leave the global collapse among the central unresolved risks for sphere recognition and to describe Section 9.1 more cautiously than its complete proof warrants.
3. This 25 August edition supersedes the 24 August report on those two points, adds the Lemma A.3 variance defect, and integrates the supplied Codex findings under an explicit evidence/provenance crosswalk. The recommendation of major revision remains unchanged.

The 24 August PDF and Markdown source are retained under `history/` rather than silently overwritten.

15 Appendix C: reproducibility

The revised review bundle contains:

- `report/revised_full_mathematical_peer_review.pdf`, the primary report;
- the corresponding Markdown and generated LaTeX sources;
- `history/full_mathematical_peer_review_2026-08-24.pdf` and `.md`;
- `audit/issue_ledger.csv` and `.json`;
- `audit/theorem_dependency_matrix.csv`;
- `audit/codex_findings_crosswalk.csv` and `.json`;
- the legacy `audit/exact_checks.py` harness and results;
- `audit/codex_integration_checks.py` and its JSON/log outputs;
- the user-supplied Codex progress log under `evidence/codex/`;
- reconstruction/source-verification evidence;
- `MANIFEST.sha256`, archive test output, and a review receipt.

To rerun both arithmetic audits from the bundle root:

```
python audit/exact_checks.py
python audit/codex_integration_checks.py
```

The report is built from Markdown with Pandoc and pdfLaTeX, rendered at 200 dpi, visually inspected, and preflighted before packaging.